

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, C++, Java, SQL, Bash

Build with Claude

Claude / Claude Code, LLM agents, tool use, automations, internal tools, integrations, prompt design, MCP, OpenAI / Groq APIs

Evaluation & Reliability

Evaluation harnesses, anti-Goodharting guardrails, holdout protection, 150+ automated tests, provider routing/fallback, instrumentation

Retrieval & RAG

RAG, RRF, cross-encoder reranking, web scraping, cell-level provenance, sentence-transformers, PyTorch

Backend & Cloud

FastAPI, Node.js, Express, REST APIs, Docker, AWS, GCP, Azure, Cloudflare Workers, Git, GitHub Actions

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Built a full-stack **internal tool** from scratch (React, Node.js, MongoDB) for the AI team at a **mental-health** company, replacing manual spreadsheet workflows with a data annotation platform used daily; cut **100+ hours/month** of manual work and improved throughput **60%**, scoping it directly with the people doing the labeling.
- Worked **cross-functionally** with clinical, product, and non-technical stakeholders to scope and translate requirements; as primary technical point of contact for the eTriage backend, ran structured knowledge transfer that onboarded another developer to on-call ownership in **~6 weeks**.
- Engineered the multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Shipped an open-source **LLM** feature to production end-to-end with **no ML mentor** on the team (4-bit quantized Llama-7B via llama.cpp, persona prompts, content-safety filter), and built 20+ Node.js + Fastify REST APIs over PostgreSQL for a customer-facing product.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables using typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

AutoEval - Self-Improving Evaluation Harness

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an evaluation-improvement agent that proposes new queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an edge-native AI research assistant: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step tasks survive process restarts and tool failures, with conversation state persisted in D1.

Community Impact & Publication

- Taught programming to 30+ underprivileged students** at Sayur, a non-profit in South Tamil Nadu, breaking technical concepts down for first-time learners.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source community projects (Linkfree, Freehit, ProjectsHut).
- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.